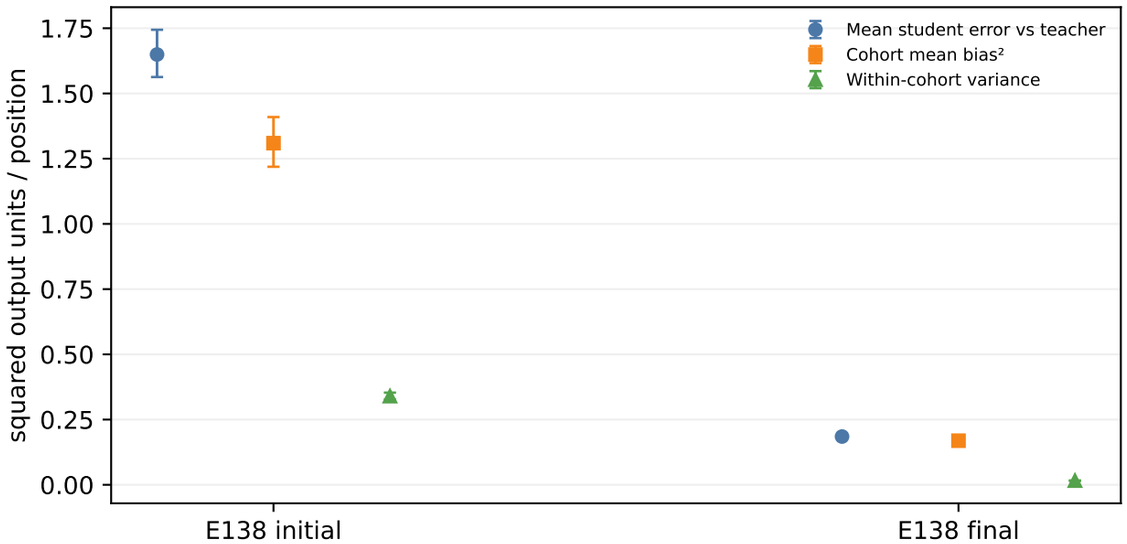
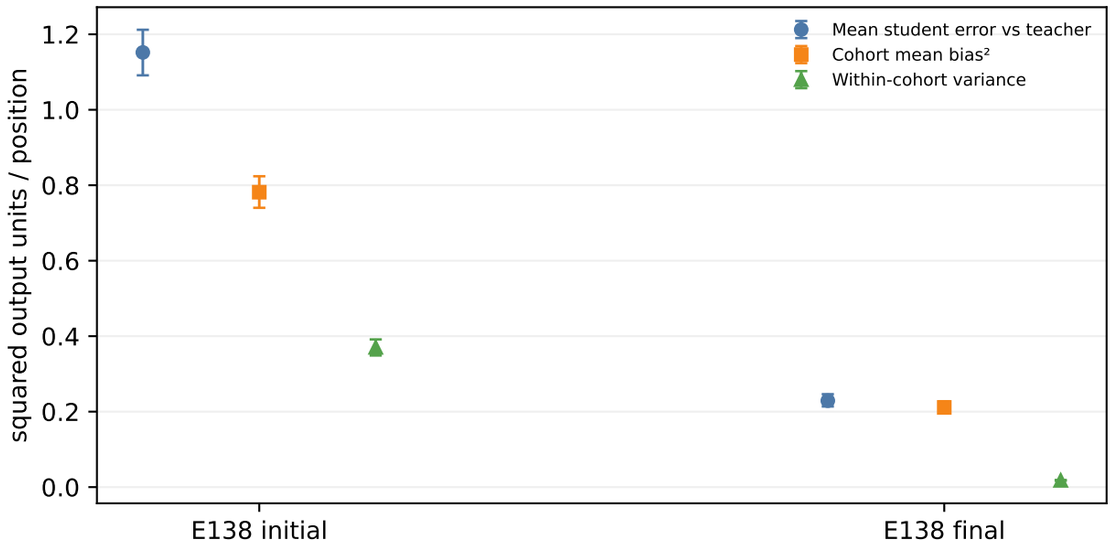
E138 — Fixed finite-cohort bias, variance and component covariance

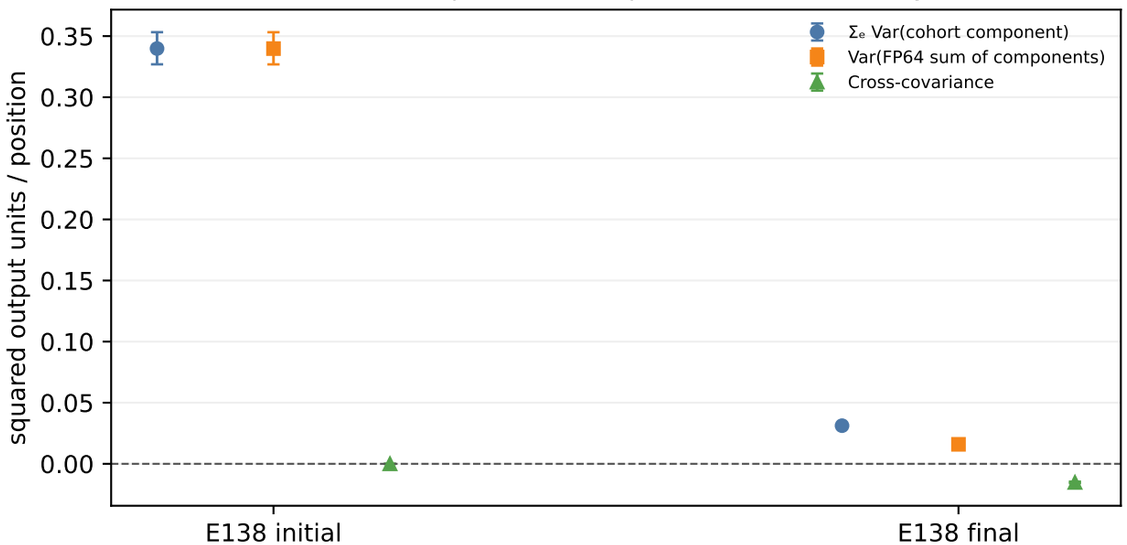
Dolma held positions: mean student teacher error = bias² + cohort variance
262,144 positions in 512 sequences; 32 file clusters



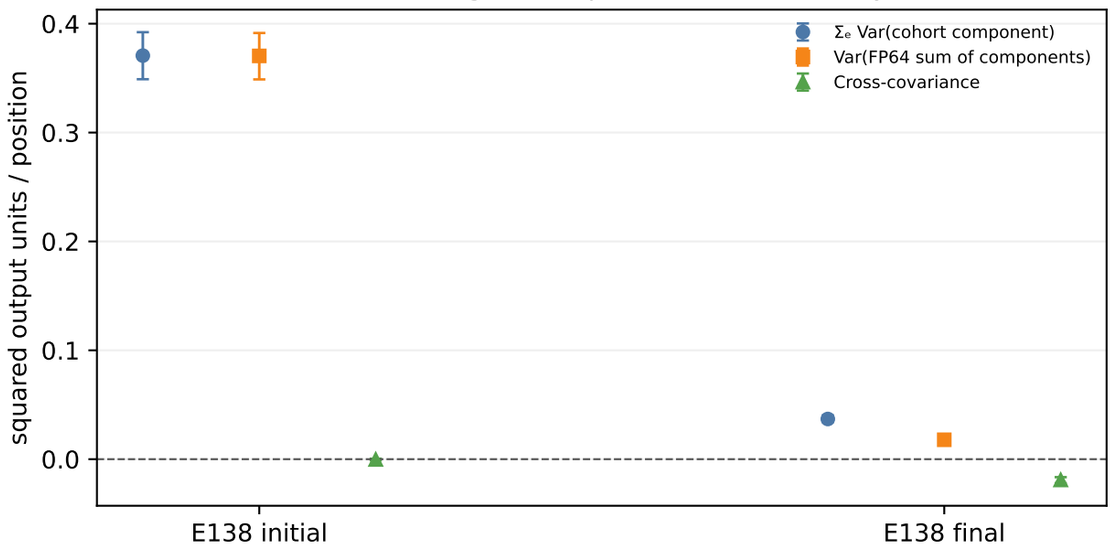
POS held targets: mean student teacher error = bias² + cohort variance
512 target positions; 48 word clusters



Dolma held positions: component variance identity



POS held targets: component variance identity



Three fixed students (seed0, reused E136 seed1, seed2); intervals resample held input clusters, not seeds or the three dependent student pairs. Each population has its own units: Dolma files vs POS words/targets. Native aggregate rounding, product rounding, and FP32−FP64 summation floors are separate in the readout table. Component covariance is an arithmetic interaction in output vectors, not semantic collaboration; no same-function or unique-role claim.